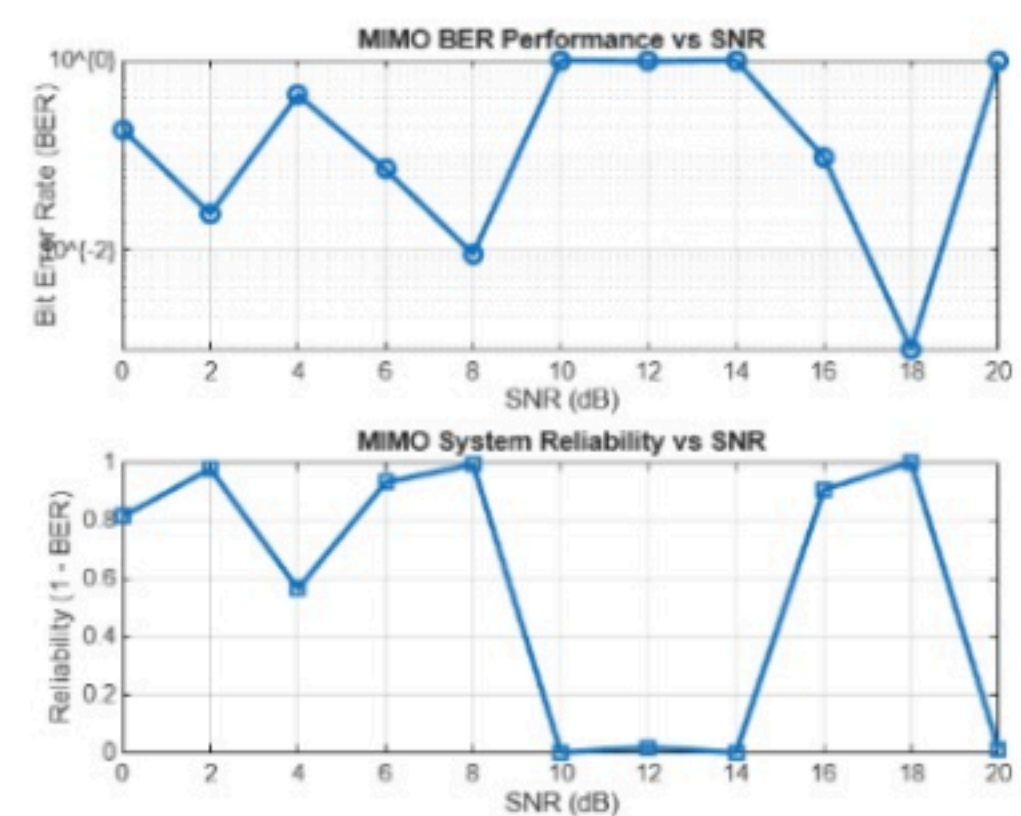


untitled4 × untitled31.m × untitled32.m × untitled7 * × untitled8 * × +

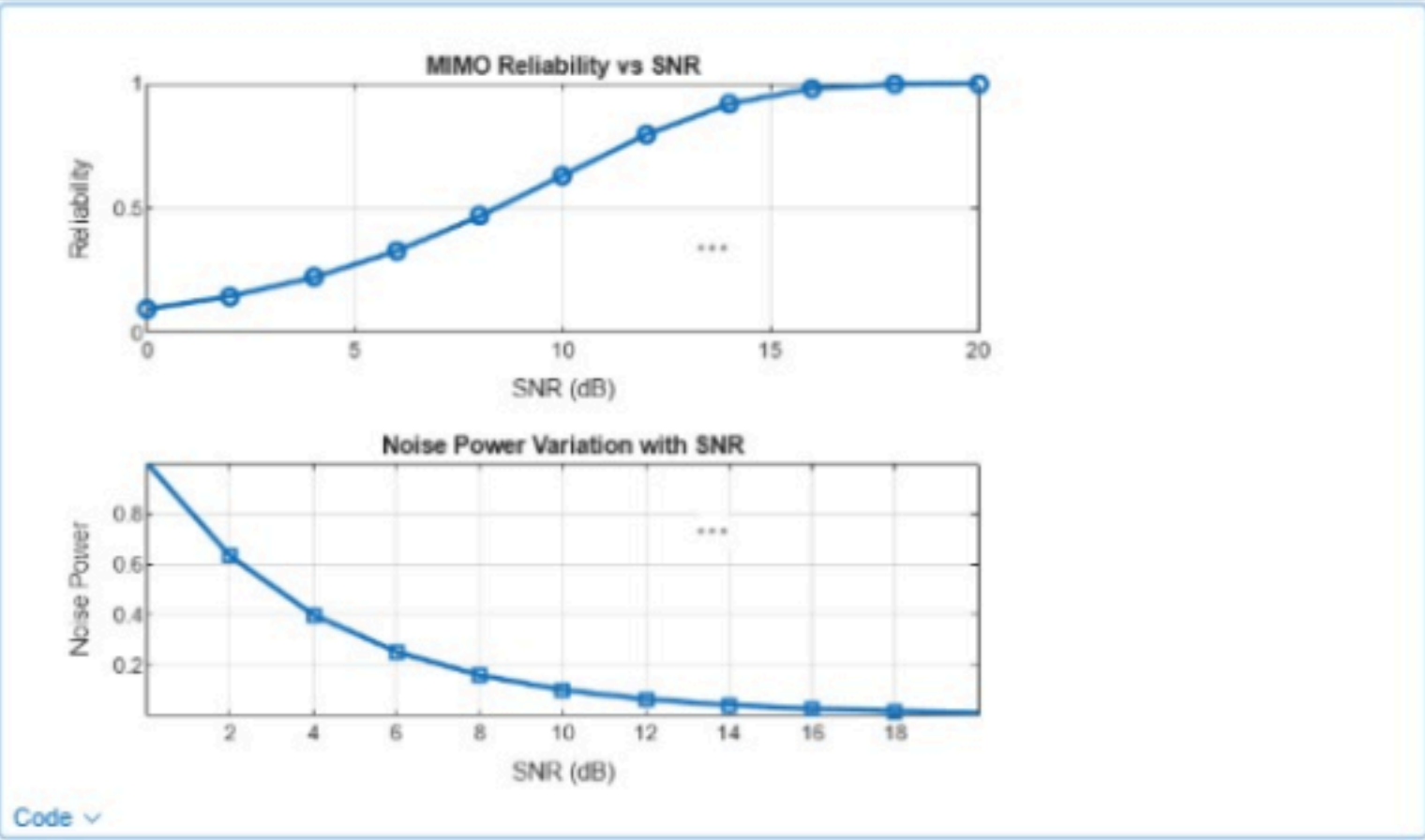
```
1 clc; clear; close all;
2 % SNR range
3 SNR_dB = 0:2:20;
4 SNR = 10.^(SNR_dB/10);
5 % MIMO parameters
6 Nt = 2; Nr = 2;
7 BER = zeros(1,length(SNR));
8 for i = 1:length(SNR)
9
10     bits = randi([0 1],10000,1);
11     symbols = 2*bits-1; % BPSK
12
13     H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
14     noise = (randn(Nr,10000) + 1j*randn(Nr,10000))/sqrt(2*SNR(i));
15
16     tx = H(1,1)*symbols' + noise(1,:);
17     rx = real(tx) > 0;
18
19     BER(i) = sum(rx' ~= bits)/length(bits);
20 end
21 figure
22 % ----- Graph 1: BER vs SNR -----
23 subplot(2,1,1)
24 semilogy(SNR_dB,BER,'-o','LineWidth',2)
25 title('MIMO BER Performance vs SNR')
26 xlabel('SNR (dB)')
27 ylabel('Bit Error Rate (BER)')
28 grid on
29 % ----- Graph 2: SNR vs Outage Probability -----
```

Command Window



```
untitled4 X untitled31.m X untitled32.m X untitled7 X untitled8 X untitled9 X
6   Nt = 2;
7   Nr = 2;
8   Reliability = zeros(1, length(SNR));
9   noise_power = zeros(1, length(SNR));
0   for i = 1:length(SNR)
1       H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
2       noise_power(i) = 1/SNR(i);
3       Reliability(i) = 1 - exp(-SNR(i)/10);
4   end
5   % ----- Graph 1: SNR vs Reliability -----
6   figure;
7   subplot(2,1,1);
8   plot(SNR_dB, Reliability, '-o', 'LineWidth', 2);
9   title('MIMO Reliability vs SNR');
0   xlabel('SNR (dB)');
1   ylabel('Reliability');
2   grid on;
3
4   % ----- Graph 2: SNR vs Noise Power -----
5   subplot(2,1,2);
6   plot(SNR_dB, noise_power, '-s', 'LineWidth', 2);
7   title('Noise Power Variation with SNR');
8   xlabel('SNR (dB)');
9   ylabel('Noise Power');
0   grid on;
1   % Display results
2   disp('SNR (dB)    Reliability    Noise Power');
3   disp([SNR_dB' Reliability' noise_power]);
4
```

Command Window



Code

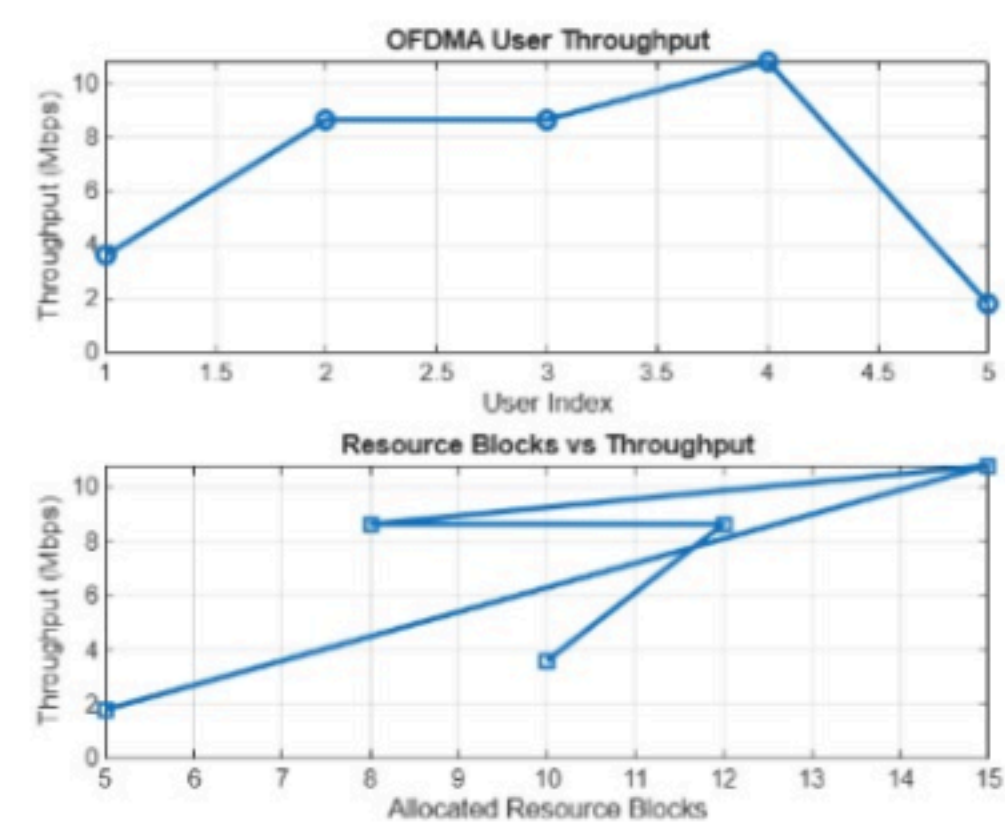
SNR (dB)	Reliability	Noise Power
0	0.0952	1.0000
2.0000	0.1466	0.6310
4.0000	0.2221	0.3981

untitled4 × untitled31.m × untitled32.m × untitled7 * × untitled8 * × untitled9 * × untitled10 * × +

```
1 clc; clear; close all;
2 rb_bw = 180e3;
3 users = 1:5;
4 alloc_RB = [10 12 8 15 5];
5 mod_eff = [2 4 6 4 2];
6 throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
7 disp('User Throughput (Mbps):')
8 disp(throughput)
9 figure
10 % ----- Graph 1: Throughput per User -----
11 subplot(2,1,1)
12 plot(users, throughput, '-o','LineWidth',2)
13 title('OFDMA User Throughput')
14 xlabel('User Index')
15 ylabel('Throughput (Mbps)')
16 grid on
17 % ----- Graph 2: RB vs Throughput -----
18 subplot(2,1,2)
19 plot(alloc_RB, throughput, '-s','LineWidth',2)
20 title('Resource Blocks vs Throughput')
21 xlabel('Allocated Resource Blocks')
22 ylabel('Throughput (Mbps)')
23 grid on
```

User Throughput (Mbps):

3.6000
8.6400
8.6400
10.8000
1.8000



Command Window